

RESEARCH ARTICLE

Doppler Effect Estimation and Compensation for Satellite Systems in Geolocation Problems

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ABSTRACT The framework of the article is a theoretical investigation of the Doppler effect on electromagnetic signals. The target application is relevant to satellite missions with resource-constrained systems based on low-cost hardware platforms and the innovation of the paper is on the error compensation strategy and on the proposal of an approximate polynomial model. In particular, the scenario of interest considers a non-cooperative emitter of electromagnetic waves on the surface of the Earth and a receiver able to perform Angle of Arrival measurements embarked on a satellite in Low Earth Orbit. Assuming the problem of geolocation is solved, and starting from the kinematics of the satellite and its relative motion with respect to the emitter, the correction on the frequency is derived with respect to the orbital parameters, which is a common representation of data for satellite systems. The purpose of the paper is not only to provide a reference model to predict the necessary correction on the frequency of the signal due to the Doppler effect, but also to present a real-time strategy for compensation in the pipeline of signal processing after its geolocation. The main goal of the paper is to give real-time estimation and compensation strategy for the Doppler factor considering only the current instantaneous kinematic information, without the information on the history of the signal over time. Furthermore, data fitted from simulations show that a first degree polynomial model has a maximum error prediction on the order of less than 0.01 percent around Nadir pointing and can be appealing for fast computation of this correction with low-cost hardware based on Field Programmable Gate Array.

INDEX TERMS Doppler effect, error compensation, signal processing, simulation, space systems.

I. INTRODUCTION

This paper addresses the challenge of accurately geolocating a non-cooperative signal emitter using a satellite platform. A primary focus is the estimation and compensation of Doppler shifts, which arise from the relative motion between the satellite and the emitter and can significantly degrade geolocation accuracy. To tackle this issue, we propose a model-based approach that exploits a satellite receiver equipped with AoA (Angle of Arrival) measurement capabilities, thereby enhancing overall positioning performance.

The associate editor coordinating the review of this manuscript and approving it for publication was Giambattista Gruosso^{ID}.

For an AoA system, a precise and accurate estimation of the frequency affects the geolocation solution in the random contribution due to the noise and the Cramér-Rao Lower Bound, [1].

A source of signal is denoted as non-cooperative when it is not supposed to be known in terms of presence, location, and transmitted message.

The measurement is performed by detecting a signal that originates from the target and travels through the space. The signal in this study is expected to consist of a collection of electromagnetic waves originating from an emitter of signal.

AoA measure consists of the direction of the signal and it is commonly referred to as LoB (Line of Bearing). In terms of technology, an array of antennas can be

used for array-beamforming to accomplish this type of measurement [2].

In particular, the study aims to address the problem of frequency correction due to the Doppler effect.

Early studies on Doppler effects in satellite systems are focused on communication links between the ground station and the satellite [3]. Its use for sensing purposes can be found in Doppler radar systems for two-way systems (i.e., the electromagnetic wave is both sent and received by the satellite); the frequency shift is then used to determine an object velocity [4]. Other studies on Doppler effects can be found in the dedicated literature on GNSS (Global Navigation Satellite Systems) [5], applications for terrestrial systems [6] or satellite-based geolocation problems [7].

Similar recent studies are on communication systems and consider circular orbit, e.g. [8] and [9]. Studies on elliptic orbits are limited to the assumption of spherical Earth and neglect relativistic effects, e.g. [10].

Other studies consider the receiver on the surface of the Earth, e.g., [11] and [12], so commonly the problem is presented in terms of azimuth and elevation, but the peculiar kinematics of the satellite is not modeled.

There are techniques to estimate Doppler shifts considering information on the variation of shape of signal over time. In general, they are based on Fourier analysis [13], Kalman filters [14], or estimators of parameters from the phase of the signal [15].

Examples of applications of signal-based techniques can be found in [16]: in particular, it is shown that for Orbcomm and NOAA (National Oceanic and Atmospheric Administration) satellite constellations the Doppler frequency can be evaluated from the frequency peaks of the cumulated Short-Time Fourier Transform of the squared signal, while for Iridium constellation, a coarse estimation is achieved through Fast Fourier Transform, while Phase Locked Loop is used to track the signal carrier frequency over time.

The error compensation of the frequency of the signal for satellite systems can be performed in two ways: the first is an offline software compensation to be performed after the data are sent from the satellite to the ground stations.

The second way is an onboard compensation of the measurement; generally, it adds computational cost on the processing unit of the satellite, and it can affect the power consumptions of the satellite.

The purpose of the paper is to give real-time estimation and compensation strategy, because the target application is the set of real-time systems.

To the knowledge of the authors, there are not attempts to estimate and correct the single measurement in AoA satellite systems with a real-time strategy and considering only instantaneous information on the kinematics available onboard of a satellite, so without the knowledge of the history and the shape of the signal.

The topic is of interest for a set of new generations and more precise technological demonstrators. One of them is SAILS (Spaceborne Autonomous Identification and

Localization System) project supervised and funded by the Italian Space Agency in which the Polytechnic University of Bari is collaborating as scientific partner. [17].

The compensation of the effect is not a new topic, but its real-time correction on resource-constrained systems, e.g., FPGA (Field Programmable Gate Array) for CubeSats platforms to present days is a challenge.

This study presents an advancement on the topic, considering the satellite kinematics on elliptic orbits, an ellipsoidal model of the Earth, and relativistic formula for the Doppler effect. Furthermore, the model is derived in order to obtain a real-time on-board strategy of compensation considering the available data on the satellite at the moment of measurement.

Another peculiarity of the system of interest, which distinguishes it from the others in literature: compared to command and telemetry subsystems, radar payloads or communication antennas, the receiver of interest is only able to detect electromagnetic waves, but it does not generate and send them.

Furthermore, the study not only presents a mathematical model for the correction of the Doppler effect, but it also provides an approximate polynomial model obtained from simulated missions in LEO (Low Earth Orbit), useful for real-time computation on low-cost hardware platforms and CubeSats [18].

In particular, the derived correction is based on the AoA measurement and the state of the satellite, which can be revealed by navigation information from the ADCD (Attitude Determination and Control System) subsystem of the satellite which is composed of a set of sensors such as accelerometers, gyroscopes, magnetometers, and star trackers [19], [20].

Among the limitations of the study, the paper does not consider other sources of change in frequency such as phenomena of scattering [21].

Since the correction of the signal due to the Doppler effect is supposed to happen after its detection, the signal is supposed to be always detected, and beyond the study there is also the problem of signal detection.

Phenomena of reflection, multipath, attenuation, and interference are not considered and are supposed to be absent.

The Doppler effect is not the only phenomenon that leads to a change in frequency for the signal: the presence of the atmosphere produces the bending of electromagnetic waves. The Doppler effect depends on the velocities of the transmitter and/or the receiver. The bending occurs when the electromagnetic wave propagates through a medium with varying refractive index (e.g., the atmosphere). The paper focuses only on the Doppler effect, and the phenomenon of the bending of electromagnetic waves is not considered among the purposes of the study.

Section II describes the mathematical model of the Doppler factor, positions and velocities, and an input-output analysis useful for software and system design. Section III presents the results of the simulations and an evaluation of a first-degree polynomial fit. In the section IV there are the conclusions of the study.

II. MATHEMATICAL MODEL

A. DOPPLER FACTOR

Doppler effect is the apparent shift in the wavelength and in the frequency of an electromagnetic wave due to the relative movement of the source of the wave and the receiver [22].

Denoting the observed frequency of the signal with f_o and the effective frequency of the signal with f_e , the Doppler effect can be calculated using (1), where $D(\Delta v)$ is the Doppler factor, which depends on the component of the relative velocity along the line from the receiver to the emitter.

$$f_e = D(\Delta v) f_o \quad (1)$$

It can be demonstrated [22] that $D(\Delta v)$ is equal to the expression in (2), where $\gamma = (1 - \beta^2)^{-1/2}$ is the Lorentz factor, $\beta = \Delta v/c$ is the ratio of velocities, and c is the speed of light.

$$D(\Delta v) = \gamma (1 + \beta) \quad (2)$$

Expressing the Lorentz factor and the ratio of velocities, the equation in (3) can be obtained.

$$D(\Delta v) = \frac{(1 + \beta)}{\sqrt{1 - \beta^2}} = \frac{1 + \frac{\Delta v}{c}}{\sqrt{1 - \left(\frac{\Delta v}{c}\right)^2}} \quad (3)$$

The component of the difference of velocities in (3) can be calculated considering the projection of the difference of velocity vectors on the line from the satellite receiver position S to the emitter position E .

Denoting with $\mathbf{v}_E$ the velocity vector of the emitter and with $\mathbf{v}_S$ the velocity vector of the satellite receiver, the component of the difference of velocity vectors can be calculated by considering the dot product in (4).

$$\Delta v = (\mathbf{v}_E - \mathbf{v}_S) \cdot \frac{\vec{SE}}{\|\vec{SE}\|} \quad (4)$$

The value of Δv in (4) depends on the positions and velocities of the satellite and the emitter, and they have to be expressed in the same coordinate system.

B. POSITION MODELS

Denoting by O the origin of a convenient coordinate system, the position vector $\vec{SE}$ can be calculated considering (5).

$$\begin{aligned} \vec{SE} &= (E - S) = (E - O) - (S - O) \\ &= \vec{OE} - \vec{OS} = (\mathbf{x}_E - \mathbf{x}_O) - (\mathbf{x}_S - \mathbf{x}_O) \end{aligned} \quad (5)$$

Considering the ECI (Earth Centered Inertial) coordinate system, (5) can be rewritten as in (6).

$$\vec{SE}|_{\text{ECI}} = \mathbf{x}_E|_{\text{ECI}} - \mathbf{x}_S|_{\text{ECI}} \quad (6)$$

Satellite position is generally expressed in terms of orbital parameters [23]: the semi-major axis a , the eccentricity of the orbit e , the inclination of the orbit i , the right ascension of the ascending node Ω , the argument of the perigee ω ,

and the true anomaly $\nu(t)$, which is the only parameter that depends on time t .

A representation of the satellite position $\mathbf{x}_S$ is in the PQW (perifocal coordinate system), as shown in (7).

$$\mathbf{x}_S|_{\text{PQW}} = \frac{a(1 - e^2)}{1 + e \cos(\nu)} \begin{pmatrix} \cos(\nu(t)) \\ \sin(\nu(t)) \\ 0 \end{pmatrix} \quad (7)$$

The conversion of data from the PQW coordinate system into the ECI coordinate system occurs through three rotations, as shown in (8), where $\mathbf{R}$ represents the matrix of rotation of the angle in the first argument about the axis of the second argument.

$$\mathbf{x}_S|_{\text{ECI}} = \mathbf{R}_{(\Omega,3)} \mathbf{R}_{(i,1)} \mathbf{R}_{(\omega,3)} \mathbf{x}_S|_{\text{PQW}} \quad (8)$$

In particular [24], the three matrices of rotations $\mathbf{R}_{(\Omega,3)}$, $\mathbf{R}_{(i,1)}$, and $\mathbf{R}_{(\omega,3)}$ are reported in (9).

$$\begin{aligned} \mathbf{R}_{(\omega,3)} &= \begin{bmatrix} \cos(\omega) & -\sin(\omega) & 0 \\ \sin(\omega) & \cos(\omega) & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \mathbf{R}_{(i,1)} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(i) & -\sin(i) \\ 0 & \sin(i) & \cos(i) \end{bmatrix} \\ \mathbf{R}_{(\Omega,3)} &= \begin{bmatrix} \cos(\Omega) & -\sin(\Omega) & 0 \\ \sin(\Omega) & \cos(\Omega) & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned} \quad (9)$$

A common representation for the emitter position is through geodetic latitude Λ , longitude λ , and height h (LLH reference system), as shown in (10).

$$\mathbf{x}_E|_{\text{LLH}} = \begin{pmatrix} \Lambda \\ \lambda \\ h \end{pmatrix} \quad (10)$$

For a receiver able to perform AoA measurements, the estimation of the emitter position on the surface of the Earth $\hat{\mathbf{x}}_E$ can be performed considering the intersection of the LoB $\mathbf{u}_{\text{LoB}}$ with the ellipsoid of the Earth. Denoting the position of the satellite S with $\mathbf{x}_S$, then $\mathbf{u}_{\text{LoB}} = (\mathbf{x}_E - \mathbf{x}_S) / \|\mathbf{x}_E - \mathbf{x}_S\|$.

The representation of the emitter position in ECI can be done considering an intermediate coordinate system ECEF (Earth Centered Earth Fixed).

Data conversion from LLH to ECEF is nonlinear [25] and, for an ellipsoidal model of the Earth, it is given by (11), where $N(\Lambda)$ is defined as in (12); $e_\oplus$ is the eccentricity of the ellipsoid and $a_\oplus$ is the ellipsoidal equatorial radius of the Earth.

For a WGS84 model [26], which has been considered in this study, $e_\oplus^2 = 0.00669437999$ and $a_\oplus = 6378.137$ km.

$$\mathbf{x}_E|_{\text{ECEF}} = \begin{pmatrix} (N(\Lambda) + h) \cos(\Lambda) \cos(\lambda) \\ (N(\Lambda) + h) \cos(\Lambda) \sin(\lambda) \\ (N(\Lambda) (1 - e_\oplus^2) + h) \sin(\Lambda) \end{pmatrix} \quad (11)$$

$$N(\Lambda) = \frac{a_\oplus}{1 - e_\oplus^2 \sin^2(\Lambda)} \quad (12)$$

Data conversion from ECEF to ECI is reported in (13), where $\alpha_{GR}(t)$ is the position of the Greenwich meridian and it is a function of time t .

$$\mathbf{x}|_{ECEF} = \mathbf{R}_{(\alpha_{GR}(t), 3)} \mathbf{x}|_{ECI} \quad (13)$$

C. VELOCITY MODELS

The velocity of a satellite in an elliptical orbit [23] represented in the PQW coordinate system is given by (14), where $\mu_{\oplus}$ is the standard gravitational parameter of the Earth equal to $3.986005 \cdot 10^{14} \text{ m}^3/\text{s}^2$.

$$\mathbf{v}_S|_{PQW} = \sqrt{\frac{\mu_{\oplus}}{a(1-e^2)}} \begin{pmatrix} -\sin(v(t)) \\ e + \cos(v(t)) \\ 0 \end{pmatrix} \quad (14)$$

The conversion of the satellite velocity $\mathbf{v}_S$ from PQW to ECI is done through the same rotations described in subsection II-B for the position, i.e., (8) and (9), and it is reported in (15).

$$\mathbf{v}_S|_{ECI} = \mathbf{R}_{(\Omega, 3)} \mathbf{R}_{(i, 1)} \mathbf{R}_{(\omega, 3)} \mathbf{v}_S|_{PQW} \quad (15)$$

Under the assumption that the emitter is fixed on the surface of the Earth, the velocity of the emitter $\mathbf{v}_E$ depends only on the angular velocity of the rotation of the Earth $\Omega_{\oplus}$, which in ECI has only a component along the z axis: $\Omega_{z\oplus}$.

The velocity of the emitter $\mathbf{v}_E$ can be evaluated through the cross product between the angular velocity of the Earth and the position of the emitter, as shown in (16).

$$\begin{aligned} \mathbf{v}_E|_{ECI} &= \Omega_{\oplus}|_{ECI} \times \vec{OE}|_{ECI} = \Omega_{\oplus}|_{ECI} \times \mathbf{x}_E|_{ECI} \\ &= \begin{pmatrix} 0 \\ 0 \\ \Omega_{z\oplus} \end{pmatrix} \times \begin{pmatrix} x_E|_{ECI} \\ y_E|_{ECI} \\ z_E|_{ECI} \end{pmatrix} = \begin{pmatrix} -\Omega_{z\oplus} y_E|_{ECI} \\ \Omega_{z\oplus} x_E|_{ECI} \\ 0 \end{pmatrix} \end{aligned} \quad (16)$$

D. INPUT-OUTPUT ANALYSIS

This subsection presents an input-output analysis that can be useful for software and system design.

Considering (4) and (6) in order to obtain the value of Δv the evaluation of four dot products, $\mathbf{v}_S \cdot \mathbf{x}_E$, $\mathbf{v}_S \cdot \mathbf{x}_S$, $\mathbf{v}_E \cdot \mathbf{x}_E$, $\mathbf{v}_E \cdot \mathbf{x}_S$, and the norm $\|\mathbf{x}_E - \mathbf{x}_S\|$ have to be performed, so the functional dependence is as shown in (17).

$$\Delta v = f(\mathbf{x}_S, \mathbf{x}_E, \mathbf{v}_S, \mathbf{v}_E) \quad (17)$$

Considering (7), (8), and (9), the dependence of $\mathbf{x}_S$ is as reported in (18).

$$\mathbf{x}_S = f(a, e, i, \Omega, \omega, v(t)) \quad (18)$$

Considering (10), (11), (12) and (13), the dependence of $\mathbf{x}_E$ is as reported in (19).

$$\mathbf{x}_E = f(\Lambda, \lambda, h, \alpha_{GR}(t)) \quad (19)$$

Considering (14), (15), and (9), the dependence of $\mathbf{v}_S$ is as reported in (20).

$$\mathbf{v}_S = f(a, e, i, \Omega, \omega, v(t)) \quad (20)$$

Considering (16) and all the considerations done for (19), the dependence of $\mathbf{v}_E$ is as reported in (21).

$$\mathbf{v}_E = f(\Lambda, \lambda, h, \alpha_{GR}(t)) \quad (21)$$

At this point it is possible to reconstruct the overall dependence of Δv , as shown in (22).

$$\Delta v = f(\Lambda, \lambda, h, \alpha_{GR}(t), a, e, i, \Omega, \omega, v(t)) \quad (22)$$

For a satellite in LEO, considering the available data onboard, only a, e, i, Ω, ω , and $v(t)$ are directly available, as they are representative of the current state of the satellite at time t . The value of $\alpha_{GR}(t)$, depending only on the time t , is also a known quantity onboard the satellite.

Conversely, the position of a noncooperative emitter is not directly known by the satellite. However, its position can be estimated in terms of geodetic latitude $\hat{\Lambda}$ and longitude $\hat{\lambda}$ by solving the geolocation problem with a single measurement; $\hat{h}$, can be obtained furthermore through multiple platforms or with multiple measurements over time with a single receiver. Studies on this kind of estimation are beyond the purpose of this article, but they have already been addressed by the authors in [27] with Newtonian mechanics and [28] with a relativistic approach.

Generally, the geolocation procedure is iterative and can be performed autonomously onboard in the case of AoA measurements. In this study, the position of the emitter is supposed to be already calculated and known.

Also beyond the scope of this article are considerations of detection performances and noise. For simplicity, the signal is supposed to always be detected, so the signal-to-noise ratio is considered $SNR \rightarrow \infty$.

Beyond the purposes of the paper is the effect of the bending of the electromagnetic waves on the frequency that comes from variations in refractive index along the path. This effect can be studied, simulated, and compensated separately, so in this paper the index of refraction is supposed to be $n = 1$. A recommendation of the ITU (International Telecommunication Union) on the topic is [29].

Considering all these factors, in principle, the expression for Δv in (22) and also the value of the Doppler factor $D(\Delta v)$ in (17) can be completely evaluated on-board and without external aids. An example of a possible scheme of computation is represented in Fig. 1.

It has to be remarked that this is a first level of estimation and compensation strategy because the Doppler shift is obtained from the AoA measurement. However, also the AoA measurement depends on the value of the frequency (in particular, in the random contribution, noise). It is not excluded that a possible two-step iterative approach can be exploited (i.e., from the AoA measurement the new frequency is predicted, then the AoA is corrected with the new value of the frequency), but its feasibility and effectiveness are beyond the purposes of the study and can be considered for future investigations.

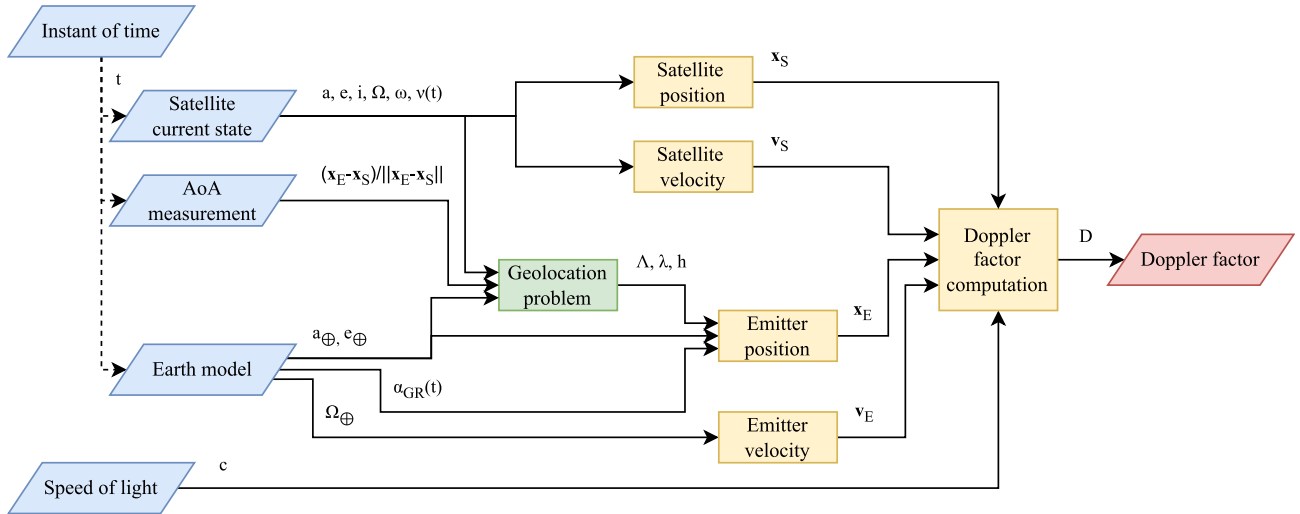


FIGURE 1. Example of computation of doppler factor D in case of a satellite in LEO and an emitter of signal on the surface of the Earth. In this study the geolocation problem, represented in green, is assumed to be solved: a possible model for its estimation in real-time is reported in [27].

For demonstrative purposes shown in subsection III, it is assumed that the emitter is on the surface of the Earth with $h = 0$.

Furthermore, and only for evaluative purposes in the simulation environment, a convenient instant of time t such that $\alpha_{GR} = 0$ is considered. Since this variable is representative of the angular position of the Greenwich meridian, this assumption is equivalent to a shift in the value of the longitude for the emitter of the signal, which is not relevant for the purposes of the study.

The value of the true anomaly $\nu(t)$ is assumed to be known and calculated by the available data from the navigation subsystem at the moment of measurement t .

Considering these last assumptions, the models in (23) and (24) can be written and considered for the simulations in section III.

$$\Delta\nu = f(\Lambda, \lambda, a, e, i, \Omega, \omega, \nu) \quad (23)$$

$$D(\Delta\nu) = D(\Lambda, \lambda, a, e, i, \Omega, \omega, \nu) \quad (24)$$

III. SIMULATIONS

In order to simulate the scenario, the software environment should be able to perform the kinematics of the orbits of the satellites.

High-level solvers of electromagnetic waves could be employed, but care must be taken to isolate only the phenomenon of the Doppler effect and not the competing effect due to the bending of electromagnetic waves.

Furthermore, a post-process module with the capability of fitting data is required.

Simulations have been conducted in the MATLAB environment, because it gives the capability to simulate all the required operations in a single environment and to avoid import/export compatibility of data.

The simulations considered the following set of input data:

- geodetic latitude of the emitter $\Lambda \in [-15, 15]^\circ$,

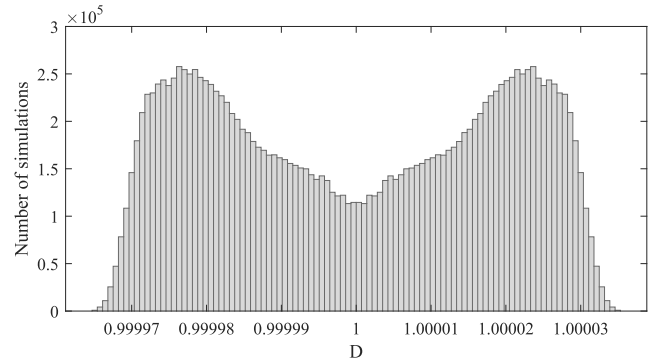


FIGURE 2. Histogram of the Doppler factor D in case of a satellite in LEO and an emitter of signal on the surface of the Earth.

- longitude of the emitter $\lambda \in [-15, 15]^\circ$,
- semi-major axis of the orbit $a \in [6671, 8171]$ km,
- eccentricity of the orbit $e \in [0, 0.3]$,
- inclination of the orbit $i \in [75, 105]^\circ$,
- right ascension of the ascending node $\Omega \in [-15, 15]^\circ$,
- argument of perigee $\omega \in [-15, 15]^\circ$,
- true anomaly $\nu \in [-15, 15]^\circ$.

For every variable 9 evenly spaced values have been considered, for a total number of simulations of 43046721.

The histogram of the values of the Doppler factor is shown in Fig. 2.

Considering that the variations of D occur in very small figures, the data are conveniently represented in terms of the red shift z , which is defined as in (25).

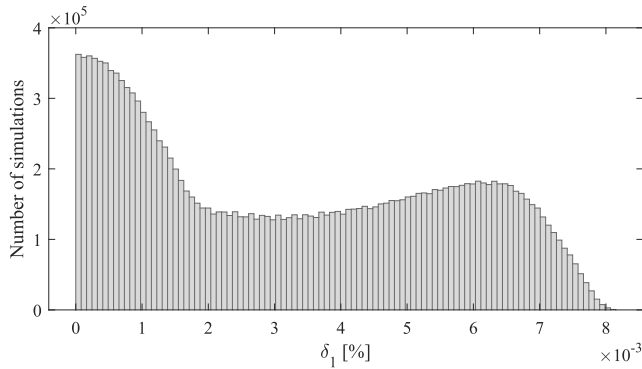
$$z = D - 1 \quad (25)$$

The maximum value of the red shift z is $3.52474 \cdot 10^{-5}$ and the minimum value is $-3.52462 \cdot 10^{-5}$.

To illustrate the effect of this Doppler shift, if an X-band radar of 10 GHz is the emitter of the signal, in the selected

TABLE 1. Values of the coefficients for the polynomial model D_1 .

Coefficient	Value
k_1	$3.04616 \cdot 10^{-6}$
k_2	$-1.80155 \cdot 10^{-6}$
k_3	$1.14970 \cdot 10^{-11}$
k_4	$3.08047 \cdot 10^{-5}$
k_5	$1.32067 \cdot 10^{-10}$
k_6	$-4.63610 \cdot 10^{-11}$
k_7	$1.80156 \cdot 10^{-6}$
k_8	$-3.11687 \cdot 10^{-5}$
k_9	$9.99998 \cdot 10^{-1}$

**FIGURE 3.** Histogram of the percentage difference of the Doppler factor δ_1 in case of a satellite in LEO and an emitter of signal on the surface of the Earth.

ranges of relative geometries between the emitter and the receiver satellite, the possible Doppler effect can produce a shift in frequency up to ± 352 kHz.

Data have been fitted in a first degree polynomial model D_1 , as reported in (26).

$$\begin{aligned}
 D(\Lambda, \lambda, a, e, i, \Omega, \omega, \nu) &\simeq D_1(\bar{\Lambda}, \bar{\lambda}, \bar{a}, \bar{e}, \bar{i}, \bar{\Omega}, \bar{\omega}, \bar{\nu}) \\
 &= k_1 \bar{\Lambda} + k_2 \bar{\lambda} + k_3 \bar{a} + k_4 \bar{e} + k_5 \bar{i} \\
 &\quad + k_6 \bar{\Omega} + k_7 \bar{\omega} + k_8 \bar{\nu} + k_9 \quad (26)
 \end{aligned}$$

In particular, the values of the coefficients $k_1, k_2, \dots, k_9$ are related to the set of normalized variables as defined in (27), in which a ‘minmax’ normalization $\bar{w}$ of a variable w is reported.

This operation guarantees both the dimensionlessness of the variables and their scaling, because the domain is transformed into a hypercube, where every variable is scaled from original values into the range from 0 to 1.

$$\bar{w}_j = \frac{w_j - \min_l(w_l)}{\max_l(w_l) - \min_l(w_l)} \quad (27)$$

The values of the coefficients of the polynomial model D_1 are reported in Table 1.

In order to evaluate the goodness of the first degree polynomial model D_1 , the percentage difference with respect to the analytical value D has been considered, as shown

in (28), which compares the absolute difference to the mean value.

$$\delta_1 = 100 \frac{|D - D_1|}{\left(\frac{D + D_1}{2}\right)} \quad (28)$$

The histogram of the values of the percentage difference δ_1 is reported in Fig. 3, the maximum value resulted in $8.14434 \cdot 10^{-3} \%$ and the mean value resulted in $3.27606 \cdot 10^{-3} \%$.

IV. CONCLUSION

This paper represents a first level of theoretical investigation of the evaluation and compensation of the Doppler effect for electromagnetic signals in the case of a noncooperative emitter of signal and a receiver able to perform AoA measurements on a satellite in LEO.

In the paper a theoretical model for the prediction of such effect is presented considering the possible available data onboard a satellite in these kinds of missions.

For real applications, other competing phenomena, such as the bending of electromagnetic waves, have to be taken into account.

Furthermore, a first-degree approximate polynomial model has been calculated from simulations. It showed an order of magnitude of less than 0.01 percent compared to analytical calculation, which is appealing for fast evaluation and compensation enabling advanced signal processing on miniaturized satellite platforms with low-cost hardware.

Future experimental tests can prove both the feasibility of the implementation of the error correction strategy and its effectiveness on CubeSat platforms equipped with FPGA.

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